Lab – Naming Docker Containers

To Begin with the Lab

Summary

In this lab, we learn how to assign custom names to Docker containers instead of relying on Docker-generated random names or container IDs. By using the --name flag with the docker run command, we can assign meaningful names to containers. These names make it easier to manage, start, stop, and communicate between containers. Container names provide a consistent way to reference containers even when their container IDs change.

Understanding Docker Container Naming

When we create a Docker container using the docker run command, Docker automatically assigns:

- A unique Container ID.
- A randomly generated container name (if we don't specify one).

However, we can assign our own custom container name using the --name flag.

Why do we use container names?

1. **Easy Management:** We can use container names instead of long container IDs.
2. **Easy Identification:** Meaningful names help identify the purpose of each container.
3. **Container Communication:** Containers can reference each other using their names when communicating over a shared Docker network.
4. **Consistency:** Custom names provide a predictable way to reference containers.

Example:

Suppose we have a Docker image named static-app.

We can create a container with a custom name:

docker run -d --name static-app-instance static-app

Here:

- docker run – Creates a new container.
- -d – Runs the container in detached mode.
- --name – Assigns a custom name to the container.
- static-app-instance – Custom name of the container.
- static-app – Name of the Docker image.

Now, instead of using the container ID, we can use static-app-instance in Docker commands.

Lab Steps

Step 1: Check Available Docker Images

First, list the available Docker images.

```
docker images
```



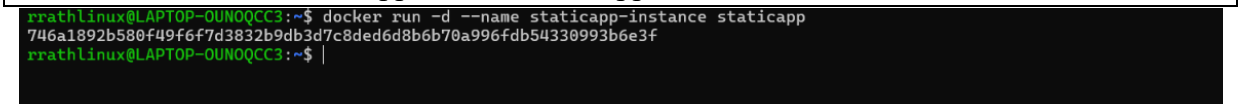
Explanation:

- Displays all locally available Docker images.
- Identify the image you want to use to create a container.

Step 2: Create a Container with a Custom Name

Use the --name flag to assign a custom name.

```
docker run -d --name staticapp-instance staticapp
```



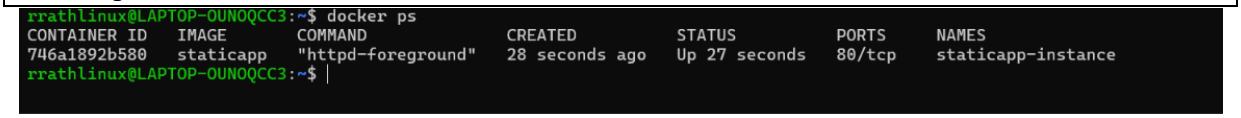
Explanation:

- Creates a container from the static-app image.
- Assigns the name staticapp-instance.
- Runs the container in detached mode.

Step 3: Verify the Container Name

Execute:

```
docker ps
```



Explanation:

This command displays running containers.

You can see the following information:

- Container ID
- Image

- Command
- Status
- Ports
- Names

The NAMES column displays your custom container name:

staticapp-instance

Step 4: Stop the Container Using Its Name

We don't need to use the container ID to stop the container.

Execute:

```
docker stop staticapp-instance
```



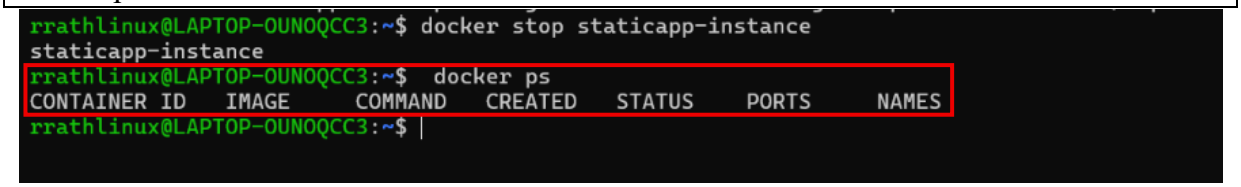
Explanation:

- docker stop – Stops a running container.
- staticapp-instance – Specifies the container by its custom name.

Step 5: Verify the Container Status

Execute:

```
docker ps
```



The stopped container will not appear because docker ps displays only running containers.

To display all containers, including stopped ones, execute:

```
docker ps -a
```

You can now see the container with its stopped status.

